

INTRODUCTION

- PROBLEM STATEMENT: Real-Time Supply-Chain Visibility and ETA Predictions
- TEAM NAME: INNOVATORS_X
- TEAM LEADER: MEGHNA MISHRA
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Project Overview – GreenRoute AI

What is GreenRoute AI?

- AI-powered real-time logistics intelligence platform
- Continuously tracks fleet movement using live GPS streams
- Predicts highly accurate ETAs using historical + live data
- Detects route deviations and transit anomalies instantly

Our Mission

- Eliminate delivery uncertainty
- Reduce fuel wastage & operational inefficiencies
- Bring transparency and intelligence to supply chains

Why It Matters

- Logistics delays cost companies millions annually
- Inaccurate ETAs reduce customer trust
- No proactive system to detect issues before they escalate

Vision

- To build a smarter, greener, and fully transparent transportation ecosystem powered by AI.

PROPOSED SOLUTION:

Green-Route AI is an intelligent route optimization and fuel efficiency monitoring system that:

- Collects real-time vehicle data
- Analyzes fuel levels and route conditions
- Optimizes path selection
- Reduces fuel consumption
- Minimizes environmental impact
- Our system uses AI-based decision logic to recommend the most efficient and eco-friendly route.



KEY FEATURES AND INNOVATION:

Key Features :

- **Real-Time Route Optimization:** AI engine selects the most fuel-efficient and fastest routes.
- **IoT-Enabled Fleet Tracking:** GPS and vehicle sensors provide continuous telematics data.
- **Adaptive to Traffic & Road Conditions:** Dynamic adjustments based on live traffic, weather, and road events.
- **Eco-Friendly Logistics:** Reduces fuel consumption and carbon emissions.
- **Multi-Modal Support:** Compatible with trucks, vans, EVs, and hybrid fleets.
- **Dashboard & Analytics:** Monitors fleet performance, fuel usage, and environmental impact.

Innovations :

- **AI-Powered Predictive Analytics:** Forecasts congestion, delivery delays, and energy consumption.
- **Continuous Self-Learning Engine:** Improves route optimization over time with more data.
- **Integration with Smart City Infrastructure:** Traffic signals, road sensors, and urban data.
- **Sustainability & ESG Reporting:** Automated carbon footprint and compliance tracking.
- **Scalable Platform:** Adaptable to fleets of any size and multiple geographic regions.



TECHNICAL COMPONENTS:

- **Frontend Layer**
- **React.js** – Builds dynamic, real-time dashboards for fleet monitoring and tracking.
- **Tailwind CSS** – Ensures responsive, modern UI with consistent visual design.
- **Axios** – Enables seamless API communication between client and server for reliable data exchange.
- **Backend Layer**
- **Node.js** – Handles real-time data processing and core system logic for operational efficiency.
- **Express.js** – Builds RESTful APIs for route tracking and ETA prediction with scalable architecture.
- **Environment Management** – Uses **CORS** and **dotenv** for secure configuration and system protection.
- **AI & Intelligence Layer**
- **OpenAI API Integration** – Provides intelligent insights and predictive assistance for strategic decision-making.
- **Custom ETA Prediction Model** – Dynamically calculates arrival times using route efficiency and data analysis.

FEASIBILITY AND VIABILITY:

Feasibility – GreenRoute AI



Technical Feasibility

- Integration with vehicle sensors & logistics software.
- Real-time route optimization capability.



Operational Feasibility

- Easy adoption by fleet managers & drivers.
- Minimal training required.



Economic Feasibility

- Budget for hardware, software & AI development.
- Cost-effective implementation.



Schedule Feasibility

- Developed & deployed within timeline.
- Agile development process.

Viability – GreenRoute AI

**Viable &
Sustainable?**

• Market Viability

- High demand from logistics companies.
- Supports eco-friendly initiatives.

• Financial Viability

- Revenue through subscriptions.
- Cost reduction in fuel & operations.

• Strategic Viability

- Aligns with sustainability goals.
- Boosts competitive advantage.

• Long-Term Sustainability

- Scalable & adaptable system.
- Future-ready for growth.

FUTURE SCOPE AND ADVANCEMENTS:

- **Smart City Integration:** Connect with traffic lights, sensors, and urban data.
- **Predictive Analytics:** Forecast congestion, delays, and fuel needs.
- **Multi-Modal Logistics:** Include EVs, drones, and hybrid transport.
- **Fleet Management Dashboards:** Real-time monitoring and intelligent suggestions.
- **EV & Renewable Integration:** Optimize routes for electric and green vehicles.
- **Scalability:** Expand to large fleets and multiple regions.
- **Sustainability Reporting:** Automated carbon emission and ESG compliance.
- **Continuous AI Learning:** Self-improving route optimization and strategies.



RESEARCH AND REFERENCES:

- AI Route Optimization:** L. Zhang & J. Wang (2022), *"AI-Driven Route Optimization for Green Logistics,"* Journal of Transport and Logistics. (Validates AI algorithms for fuel-efficient route planning)
- IoT Integration in Fleets:** R. Patel & S. Kumar (2021), *"Sustainable Fleet Management using IoT and AI,"* Int. Journal of Smart Transportation. (Shows sensor-based vehicle monitoring improves operational efficiency)
- Green Logistics Market:** McKinsey & Company (2019), *"Sustainable Mobility: How AI is Transforming Logistics,"* Industry Insights. (Demonstrates demand for eco-friendly logistics solutions)
- Financial Efficiency:** A. Gaur & P. Verma (2020), *"Cost-Benefit Analysis of AI in Logistics Operations,"* Procedia Computer Science. (Proves cost reduction and ROI from AI-enabled route optimization)
- Sustainability Impact:** International Energy Agency (IEA, 2020), *"The Future of Trucks: Implications for Energy and Environment,"* IEA Reports. (Supports environmental benefits and emission reduction)
- Scalability & Smart Cities:** L. Zhou et al. (2019), *"Intelligent Transportation Systems for Smart Cities: AI and IoT Integration,"* IEEE Trans. on Intelligent Transportation Systems. (Confirms that AI-IoT platforms can scale for urban fleets)



THANK YOU